

Report for Forest Query into Global Deforestation, 1990 to 2016

Forest Query is on a mission to combat deforestation around the world and to raise awareness about this topic and its impact on the environment. The data analysis team at ForestQuery has obtained data from the World Bank that includes forest area and total land area by country and year from 1990 to 2016, as well as a table of countries and the regions to which they belong.

The data analysis team has used SQL to bring these tables together and to query them in an effort to find areas of concern as well as areas that present an opportunity to learn from successes.

1. GLOBAL SITUATION

According to the World Bank, the total forest area of the world was **41282695 km²** in 1990. As of 2016, the most recent year for which data was available, that number had fallen to **39958246 km²**, a loss of **1324449 km²**, or **-3.21%**.

The forest area lost over this time period is slightly more than the entire land area of **Peru** listed for the year 2016 (which is **1280000 km²**)

2. REGIONAL OUTLOOK

In 2016, the percent of the total land area of the world designated as forest was **31.38%**. The region with the highest relative forestation was **Latin America & Caribbean**, with **46.16%**, and the region with the lowest relative forestation was **Middle East & North Africa** with **2.07%** forestation.

In 1990, the percent of the total land area of the world designated as forest was **32.42%**. The region with the highest relative forestation was **Latin America & Caribbean**, with **51.03%**, and the region with the lowest relative forestation was **Middle East & North Africa**, with **1.78%** forestation.

Table 2.1: Percent Forest Area by Region, 1990 & 2016:

Region	1990 Forest Percentage	2016 Forest Percentage
Latin America & Caribbean	51.03	46.16
Sub-Saharan Africa	30.67	28.79
World	32.42	31.38
Middle East & North Africa	1.78	2.07
North America	35.65	36.04
East Asia & Pacific	25.78	26.36
Europe & Central Asia	37.28	38.04
South Asia	16.51	17.51

The only regions of the world that decreased in percent forest area from 1990 to 2016 were **Latin America & Caribbean** (dropped from **51.03%** to **46.16%**) and **Sub-Saharan Africa** (**30.67%** to **28.79%**). All other regions actually increased in forest area over this time period. However, the drop in forest area in the two aforementioned regions was so large, the percent forest area of the world decreased over this time period from **32.42%** to **31.38%**.

3. COUNTRY-LEVEL DETAIL

A. SUCCESS STORIES

There is one particularly bright spot in the data at the country level, **China**. This country actually increased in forest area from 1990 to 2016 by **527229.062 km²**. It would be interesting to study what has changed in this country over this time to drive this figure in the data higher. The country with the next largest increase in forest area from 1990 to 2016 was the **United States**, but it only saw an increase of **79200 km²**, much lower than the figure for **China**.

China and **United States** are of course very large countries in total land area, so when we look at the largest *percent* change in forest area from 1990 to 2016, we aren't surprised to find a much smaller country listed at the top. **Iceland** increased in forest area by **313.66%** from 1990 to 2016.

B. LARGEST CONCERNS

Which countries are seeing deforestation to the largest degree? We can answer this question in two ways. First, we can look at the absolute square kilometer decrease in forest area from 1990 to 2016. The following 3 countries had the largest decrease in forest area over the time period under consideration:

Table 3.1: Top 5 Amount Decrease in Forest Area by Country, 1990 & 2016:

Country	Region	Absolute Forest Area Change
Brazil	Latin America & Caribbean	-541510 km ²
Indonesia	East Asia & Pacific	-282194 km ²
Myanmar	East Asia & Pacific	-107234 km ²
Nigeria	Sub-Saharan Africa	-106506 km ²
Tanzania	Sub-Saharan Africa	-102320 km ²

The second way to consider which countries are of concern is to analyze the data by percent decrease.

Table 3.2: Top 5 Percent Decrease in Forest Area by Country, 1990 & 2016:

Country	Region	Pct Forest Area Change
Togo	Sub-Saharan Africa	-75.45%
Nigeria	Sub-Saharan Africa	-61.80%
Uganda	Sub-Saharan Africa	-59.13%
Mauritania	Sub-Saharan Africa	-46.75%
Honduras	Latin America & Caribbean	-45.03%

When we consider countries that decreased in forest area percentage the most between 1990 and 2016, we find that four of the top 5 countries on the list are in the region of **Sub-Saharan Africa**. The countries are **Togo**, **Nigeria**, **Uganda**, and **Mauritania**. The 5th country on the list is **Honduras**, which is in the **Latin America & Caribbean** region.

From the above analysis, we see that **Nigeria** is the only country that ranks in the top 5 both in terms of absolute square kilometer decrease in forest as well as percent decrease in forest area

from 1990 to 2016. Therefore, this country has a significant opportunity ahead to stop the decline and hopefully spearhead remedial efforts.

C. QUARTILES

Table 3.3: Count of Countries Grouped by Forestation Percent Quartiles, 2016:

Quartile	Number of Countries
0-25%	84
25-50%	73
50-75%	38
75-100%	9

The largest number of countries in 2016 were found in the 1st quartile.

There were 9 countries in the top quartile in 2016. These are countries with a very high percentage of their land area designated as forest. The following is a list of countries and their respective forest land, denoted as a percentage.

Table 3.4: Top Quartile Countries, 2016:

Country	Region	Pct Designated as Forest
American Samoa	East Asia & Pacific	87.50
Micronesia, Fed. Sts.	East Asia & Pacific	91.86
Gabon	Sub-Saharan Africa	90.04
Guyana	Latin America & Caribbean	83.90
Lao PDR	East Asia & Pacific	82.11
Pala	East Asia & Pacific	87.61
Solomon Islands	East Asia & Pacific	77.86
Suriname	Latin America & Caribbean	98.26

Seychelles	Sub-Saharan Africa	88.41
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4. RECOMMENDATIONS

Write out a set of recommendations as an analyst on the Forest Query team.

- **What have you learned from the World Bank data?**

There are some interesting insight in this data, the only two region that have an overall decline in forestation areas were "Sub-Saharan Africa" and "Latin America & Caribbean". but if we took a closer look into these countries, from 1990 and up to 2016 we will find countries such as Cabo Verde and Eswatini in Sub-Saharan Africa region that has an increase in the forestation area by 8.2% and 6.9%, and if we looked at Latin America & Caribbean region we will find some countries that has flourished in this period such as Puerto Rico , Dominican Republic and Cuba by 23.9% , 18.8% and 12.1%. the rest of the countries in these two regions must follow these steps and study what has been done.

- **Which countries should we focus on over others?**

Will, there are many approaches that we can take to decide which countries to focus on, if we looked on the decreased in forest percentage then it's cleared that it must be the Sub-Saharan Africa region specifically Togo, Nigeria, Uganda and Mauritania. But if we looked at it from amount decreased in forest area then it should be Brazil, Indonesia, Myanmar and Nigeria.

we need to look what is the root cause of this decline in forest it could be cause by human such as commercial farming or natural disaster such as wildfires.

5. APPENDIX: SQL Queries Used

```
CREATE VIEW forestation AS(
SELECT f.country_code,
       f.country_name,
       f.year,
       f.forest_area_sqkm,
       l.total_area_sq_mi * 2.59::double precision AS land_area_sqkm,
       r.region,
       r.income_group,
       f.forest_area_sqkm / (l.total_area_sq_mi * 2.59::double precision) * 100::double precision AS forest_area
FROM forest_area f
JOIN land_area l ON f.country_code::text = l.country_code::text AND f.year = l.year
JOIN regions r ON f.country_code::text = r.country_code::text
```

```
ORDER BY f.country_code, f.year)
```

```
/*
```

```
Part 1 - Global Situation
```

```
*/
```

```
/*
```

```
a. What was the total forest area (in sq km) of the world in 1990?
```

```
Please keep in mind that you can use the country record denoted as "World" in the region table.
```

```
*/
```

```
SELECT ROUND(forest_area_sqkm) AS forest_sqkm_1990
```

```
FROM forestation
```

```
WHERE region = 'World'
```

```
AND year = 1990;
```

```
-- 41,282,695
```

```
/*
```

```
b. What was the total forest area (in sq km) of the world in 2016?
```

```
Please keep in mind that you can use the country record in the table is denoted as "World."
```

```
*/
```

```
SELECT ROUND(forest_area_sqkm) AS forest_sqkm_1990
```

```
FROM forestation
```

```
WHERE region = 'World'
```

```
AND year = 2016;
```

```
-- 39,958,246
```

```
/*
```

```
c. What was the change (in sq km) in the forest area of the world from 1990 to 2016?
```

```
*/
```

```
SELECT (ROUND(a.forest_area_sqkm) - ROUND(b.forest_area_sqkm)) AS difference
```

```
FROM forestation a
```

```
JOIN forestation b
```

```
ON a.region = b.region
```

```
WHERE a.region = 'World'
```

```
AND a.year = 1990
```

```
AND b.year=2016
```

```
-- 1,324,449
```

```
/*
```

d. What was the percent change in forest area of the world between 1990 and 2016?

```
*/
```

```
WITH cte90 AS(
```

```
    SELECT ROUND(forest_area_sqkm) AS forest_area_sqkm_90
```

```
    FROM forestation
```

```
    WHERE region='World'
```

```
    AND year = 1990
```

```
),
```

```
cte16 AS (
```

```
SELECT ROUND(forest_area_sqkm) AS forest_area_sqkm_16
```

```
    FROM forestation
```

```
    WHERE region='World'
```

```
    AND year = 2016
```

```
)
```

```
SELECT
```

```
ROUND(CAST(
```

```
(
```

```
(cte16.forest_area_sqkm_16 - cte90.forest_area_sqkm_90) / cte90.forest_area_sqkm_90
```

```
)*100 AS NUMERIC),2) || '%' AS forest_area_change
```

```
FROM cte90,cte16;
```

```
-- -3.21%
```

```
/*
```

e. If you compare the amount of forest area lost between 1990 and 2016,

to which country's total area in 2016 is it closest to?

```
*/
```

```
SELECT DISTINCT country_name,
```

```
    ROUND(land_area_sqkm) AS land_area_sqkm
```

```
FROM forestation
```

```
WHERE land_area_sqkm <=(
```

```

WITH cte90 AS(
    SELECT forest_area_sqkm
    FROM forestation
    WHERE year = 1990
    AND country_name = 'World'
),
cte16 AS(
    SELECT forest_area_sqkm
    FROM forestation
    WHERE year = 2016
    AND country_name = 'World'
)
SELECT (cte90.forest_area_sqkm - cte16.forest_area_sqkm) AS forest_area_change
FROM cte90,cte16
)
ORDER BY land_area_sqkm desc
LIMIT 1;
-- 1280000

```

```

/*

```

```

Part 2 - Regional Outlook

```

```

*/

```

```

/*

```

```

Create a table that shows the Regions and their percent forest area
(sum of forest area divided by the sum of land area) in 1990 and 2016.
(Note that 1 sq mi = 2.59 sq km).

```

```

*/

```

```

CREATE VIEW forest_to_land as(
SELECT region,
    year,
    (sum(forest_area_sqkm) / sum(land_area_sqkm)*100) forest_to_land_percent
FROM forestation
--WHERE YEAR IN (2016,1990)
GROUP BY region,year
ORDER BY year,forest_to_land_percent desc);

```



```
/*
```

a. What was the percent forest of the entire world in 2016?

Which region had the HIGHEST percent forest in 2016, and which had the LOWEST, to 2 decimal places?

```
*/
```

```
SELECT region , ROUND(CAST(forest_to_land_percent AS NUMERIC),2)
```

```
FROM forest_to_land
```

```
where year= 2016;
```

```
/*
```

b. What was the percent forest of the entire world in 1990?

Which region had the HIGHEST percent forest in 1990, and which had the LOWEST, to 2 decimal places?

```
*/
```

```
SELECT region , ROUND(CAST(forest_to_land_percent AS NUMERIC),2)
```

```
FROM forest_to_land
```

```
where year= 1990;
```

```
/*
```

c. Based on the table you created, which regions of the world DECREASED in forest area from 1990 to 2016?

```
*/
```

```
WITH t90 AS(
```

```
SELECT sub.*,
```

```
    (sub.forest_area / sub.land_area) * 100 AS forest_percentage
```

```
FROM
```

```
    (SELECT region,
```

```
        ROUND(CAST(SUM(forest_area_sqkm) AS NUMERIC),2) AS forest_area,
```

```
        ROUND(CAST(SUM(land_area_sqkm) AS NUMERIC),2) AS land_area,
```

```
        year
```

```
    FROM forestation
```

```
    GROUP BY region, year
```

```
    HAVING year =1990
```

```
) as sub
```

```
ORDER BY forest_percentage
```

```
),
```

```
t16 AS(
```

```
SELECT sub.*,
```

```

        (sub.forest_area / sub.land_area) * 100 AS forest_percentage
FROM
    (SELECT region,
        ROUND(CAST(SUM(forest_area_sqkm) AS NUMERIC),2) AS forest_area,
        ROUND(CAST(SUM(land_area_sqkm) AS NUMERIC),2) AS land_area,
        year
    FROM forestation
    GROUP BY region, year
    HAVING year =2016
    ) as sub
ORDER BY forest_percentage
)

SELECT t90.region,
    ROUND((t16.forest_percentage - t90.forest_percentage),2) AS change_percentage
FROM t90
JOIN t16
ON t90.region = t16.region
AND t90.forest_area > t16.forest_area
ORDER BY change_percentage;

```

```

/*

```

Part 3 - Country-Level Detail

```

*/

```

```

/*

```

a. Which 5 countries saw the largest amount decrease in forest area from 1990 to 2016? What was the difference in forest area for each?

```

*/

```

```

WITH t90 AS(
    SELECT country_name,
        CAST(forest_area_sqkm AS NUMERIC) AS forest90,
        forest_area AS forest_area90,
        region
    FROM forestation
    WHERE year =1990

```

```

    AND country_name <> 'World'

),

t16 AS(
    SELECT country_name,
           CAST(forest_area_sqkm AS NUMERIC) AS forest16,
           forest_area AS forest_area16,
           region
    FROM forestation
    WHERE year =2016
    AND country_name <> 'World'

)

SELECT t90.country_name,
       t90.region,
       ROUND(CAST(t90.forest90 AS NUMERIC),2) AS forest_sqkm90,
       ROUND(CAST(t16.forest16 AS NUMERIC),2) AS forest_sqkm16,
       ROUND(COALESCE((t16.forest16 / t90.forest90)*100,0),2) AS difference_Percentage,
       COALESCE((t16.forest16 - t90.forest90),0) AS difference_km

FROM t90
JOIN t16
ON t90.country_name = t16.country_name
ORDER BY difference_km
LIMIT 5;

```

/*

b. Which 5 countries saw the largest percent decrease in forest area from 1990 to 2016? What was the percent change to 2 decimal places for each?

*/

```

WITH t90 AS(
    SELECT country_name,
           CAST(forest_area_sqkm AS NUMERIC) AS forest90,
           forest_area AS forest_area90,
           region
    FROM forestation

```

```

WHERE year =1990
AND country_name <> 'World'
),
t16 AS(
SELECT country_name,
        CAST(forest_area_sqkm AS NUMERIC) AS forest16,
        forest_area AS forest_area16,
        region
FROM forestation
WHERE year =2016
AND country_name <> 'World'
)
SELECT t90.country_name,
        t90.region,
        ROUND(CAST(t90.forest90 AS NUMERIC),2) AS forest_sqkm90,
        ROUND(CAST(t16.forest16 AS NUMERIC),2) AS forest_sqkm16,
        ROUND(COALESCE(((t16.forest16 - t90.forest90)/t90.forest90)*100,0),2) AS difference_Percentage
        --COALESCE((t16.forest16 - t90.forest90),0) AS difference_km

FROM t90
JOIN t16
ON t90.country_name = t16.country_name
ORDER BY difference_Percentage
LIMIT 5;

```

```

/*

```

c. If countries were grouped by percent forestation in quartiles, which group had the most countries in it in 2016?

```

*/

```

```

WITH sub AS(
SELECT country_name,
CASE
    WHEN forest_area < 25 THEN '0-25%'
    WHEN forest_area >= 25
        AND
        forest_area < 50
    THEN '25-50%'

```

```

        WHEN forest_area >= 50
            AND
                forest_area < 75
        THEN '50-75%'
        ELSE '75-100%'
        END AS quartiles
    FROM forestation
    WHERE year = 2016
        AND
            forest_area IS NOT NULL
)
SELECT DISTINCT quartiles,
    COUNT(country_name) OVER (PARTITION BY quartiles) AS number_of_countries
FROM sub
ORDER BY quartiles ;

```

/*

d. List all of the countries that were in the 4th quartile (percent forest > 75%) in 2016.

*/

```

WITH sub AS(
    SELECT country_name,region,forest_area,
    CASE
        WHEN forest_area < 25 THEN '0-25%'
        WHEN forest_area >= 25
            AND
                forest_area < 50
        THEN '25-50%'
        WHEN forest_area >= 50
            AND
                forest_area < 75
        THEN '50-75%'
        ELSE '75-100%'
        END AS quartiles
    FROM forestation
    WHERE year = 2016
        AND
            forest_area IS NOT NULL

```

```
)  
SELECT country_name,  
       region,  
       ROUND(CAST(forest_area AS NUMERIC),2) AS percentage,  
       quartiles  
FROM sub  
WHERE quartiles = '75-100%';
```

```
/*
```

e. How many countries had a percent forestation higher than the United States in 2016?

```
*/
```

```
SELECT COUNT(*)  
FROM (  
    select DISTINCT country_name  
    FROM forestation  
    WHERE forest_area >(  
        SELECT forest_area  
        FROM forestation  
        WHERE country_name ='United States'  
        AND year =2016  
    )  
) AS sub
```